12TH CHEMISTRY · CHAPTER 21 · FULL CHAPTER NOTES

Halogenoalkanes

Topic-wise short and long questions with answers from the complete chapter

INTRODUCTION AND CLASSIFICATION

Q.1 [Short] Why are halogenoalkanes useful in daily life?

Halogenoalkanes are very useful in daily life, mainly because of their specialized chemical stability:

- In the past, chlorofluorocarbons (CFCs) were used as **refrigerants** and aerosol propellants; they are being replaced by eco-friendly alternatives such as Freons.
- In medicine, halogenated compounds serve as powerful **anesthetic** agents.
- They are vital intermediates in manufacturing **polymers**, such as PVC (polyvinyl chloride) for piping and PTFE (Teflon) for non-stick cookware.

Q.2 [Short] What is halothane? Why has it replaced chloroform?

Halothane, CF_3–CHClBr, is a halogenated hydrocarbon widely used as a powerful inhalation anesthetic. It induces a reversible loss of consciousness during surgeries. Previously, chloroform and other chlorinated anesthetics were used, which are now banned due to their toxicity and safety concerns.

Q.3 [Short] What are halogenoalkanes? What are alkyl halides?

A halogenoalkane is formed when one or more hydrogen atoms of an alkane are replaced by halogen atoms. They may be mono-, di-, tri- or polyhalogenoalkanes depending on the number of halogen atoms (–X) present in the molecule, e.g., CH_3X, CH_2X_2, CHX_3. Monohalogenoalkanes are also called **alkyl halides**. Their general formula is R–X, where R may be methyl, ethyl, propyl, etc., and X is a halogen atom (F, Cl, Br, I).

Q.4 [Long] Classify monohalogenoalkanes into primary, secondary and tertiary with examples.

Type	Halogen-bearing carbon is attached to	Examples
Primary	one or no other carbon atom	CH_3–Cl chloromethane; CH_3–CH_2–Cl chloroethane; CH_3–CH_2–CH_2–Cl 1-chloropropane
Secondary	two other carbon atoms directly	$(CH_3)_2$CH–Cl 2-chloropropane; CH_3–CHCl–CH_2–CH_3 2-chlorobutane
Tertiary	three other carbon atoms directly	$(CH_3)_3$C–Cl 2-chloro-2-methylpropane

Q.5 [Short] How does the molecular structure (primary, secondary, tertiary) affect the reactivity of a halogenoalkane?

The type of carbon bearing the halogen decides the mechanism. Primary halogenoalkanes have an uncrowded carbon and react by S_N2 (and E2). Tertiary halogenoalkanes form stable tertiary carbocations and have a crowded carbon, so they react by S_N1 (and E1). Secondary halogenoalkanes can follow both, depending on the solvent.

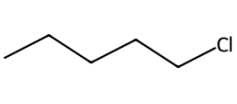
Quick Check 21.1

Q.6 [Short] a) Write down the IUPAC name of the compound, $(CH_3)_2CH\text{-}CH_2\text{-}Cl$

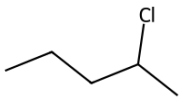
The longest chain has three carbon atoms, with chlorine on C-1 and a methyl group on C-2: **1-chloro-2-methylpropane** (a primary halogenoalkane).

Q.7 [Short] b) Draw all possible structures of a compound having the molecular formula $C_5H_{11}Cl$. Classify each as primary, secondary or tertiary chloride.

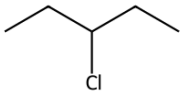
There are eight structural isomers: four primary, three secondary and one tertiary.



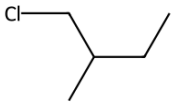
1-chloropentane (primary)



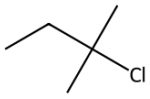
2-chloropentane (secondary)



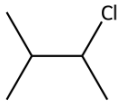
3-chloropentane (secondary)



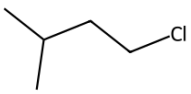
1-chloro-2-methylbutane (primary)



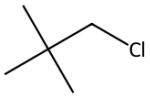
2-chloro-2-methylbutane (tertiary)



2-chloro-3-methylbutane (secondary)



1-chloro-3-methylbutane (primary)



1-chloro-2,2-dimethylpropane (primary)

Name	Condensed formula	Class
1-chloropentane	$CH_3CH_2CH_2CH_2CH_2Cl$	Primary
2-chloropentane	$CH_3CH_2CH_2CHClCH_3$	Secondary
3-chloropentane	$CH_3CH_2CHClCH_2CH_3$	Secondary
1-chloro-2-methylbutane	$CH_3CH_2CH(CH_3)CH_2Cl$	Primary
2-chloro-2-methylbutane	$CH_3CH_2CCl(CH_3)_2$	Tertiary
2-chloro-3-methylbutane	$(CH_3)_2CHCHClCH_3$	Secondary
1-chloro-3-methylbutane	$(CH_3)_2CHCH_2CH_2Cl$	Primary
1-chloro-2,2-dimethylpropane	$(CH_3)_3CCH_2Cl$	Primary

21.1 PREPARATION OF HALOGENOALKANES

Q.8 [Long] **Describe the methods of preparation of halogenoalkanes.**

i) Halogenation of alkanes in the presence of light (free radical substitution).

$$CH_3–CH_3 + Cl_2 \rightarrow CH_3–CH_2–Cl + HCl \text{ (light)}$$

ii) Electrophilic addition of hydrogen halides to alkenes

$$H_2C=CH_2 + HBr \rightarrow CH_3–CH_2–Br$$

iii) Alcohol with halogen acid in the presence of $ZnCl_2$, which acts as a catalyst.

$$CH_3–CH_2–OH + HCl \rightarrow CH_3–CH_2–Cl + H_2O\ (ZnCl_2)$$

iv) Alcohol with thionyl chloride in pyridine as a catalyst. This method is especially useful because the by-products (HCl and SO_2) are gases which go off, leaving behind the pure product.

$$CH_3–CH_2–OH + SOCl_2 \rightarrow CH_3–CH_2–Cl + SO_2 + HCl \text{ (pyridine)}$$

v) Alcohol with phosphorus trihalides or pentahalides which replace the –OH group by a halo group.

$$3CH_3–CH_2–OH + PBr_3 \rightarrow 3CH_3–CH_2–Br + H_3PO_3$$
$$CH_3–CH_2–OH + PCl_5 \rightarrow CH_3–CH_2–Cl + POCl_3 + HCl$$

vi) Alcohol with potassium bromide and an acid such as H_2SO_4 or H_3PO_4. This is commonly used to prepare bromoalkanes and iodoalkanes. The acid first reacts with KBr to give HBr, which then reacts with the alcohol. Phosphoric acid is preferred for iodoalkanes because it is a weak oxidizing agent and is less likely to oxidize iodide ions to iodine.

$$CH_3–CH_2–OH + KBr + H_2SO_4 \rightarrow CH_3–CH_2–Br + H_2O + KHSO_4$$

Q.9 [Short] **Why is the thionyl chloride method especially useful for preparing alkyl chlorides?**

Because the by-products, HCl and SO_2, are both gases. They escape from the reaction mixture, leaving behind the pure alkyl chloride.

$$CH_3–CH_2–OH + SOCl_2 \rightarrow CH_3–CH_2–Cl + SO_2 + HCl \text{ (pyridine)}$$

Q.10 [Short] **Why is phosphoric acid preferred to sulphuric acid for preparing iodoalkanes?**

Phosphoric acid is a weak oxidizing agent, so it is less likely to oxidize iodide ions to iodine. The hydrogen iodide formed is therefore available to react with the alcohol.

Q.11 [Short] **What is the role of zinc chloride in the reaction of an alcohol with HCl?**

$ZnCl_2$ acts as a catalyst for the conversion of the alcohol into the alkyl chloride.

Q.12 [Short] **How is bromoethane produced in nature, and what is its effect on the atmosphere?**

Bromoethane is naturally emitted mainly by marine algae and phytoplankton such as *Trichodesmium*. These organisms convert seawater bromide into volatile organobromides. Once in the atmosphere, bromoethane contributes to the global bromine cycle and plays a role in stratospheric ozone depletion.

Quick Check 21.2

Q.13 [Short] **a) Give two methods to convert methanol into chloromethane.**

1. With hydrogen chloride in the presence of $ZnCl_2$ as catalyst. **2.** With thionyl chloride in pyridine (PCl_5 can also be used).

$$CH_3\text{–}OH + HCl \rightarrow CH_3\text{–}Cl + H_2O\ (ZnCl_2)$$
$$CH_3\text{–}OH + SOCl_2 \rightarrow CH_3\text{–}Cl + SO_2 + HCl\ (pyridine)$$

Q.14 [Short] **b) Give reactions of propanol with $SOCl_2$ and PBr_3.**

$$CH_3\text{–}CH_2\text{–}CH_2\text{–}OH + SOCl_2 \rightarrow CH_3\text{–}CH_2\text{–}CH_2\text{–}Cl + SO_2 + HCl\ (pyridine)$$
$$3CH_3\text{–}CH_2\text{–}CH_2\text{–}OH + PBr_3 \rightarrow 3CH_3\text{–}CH_2\text{–}CH_2\text{–}Br + H_3PO_3$$

The products are 1-chloropropane and 1-bromopropane.

Q.15 [Short] **c) Can we prepare fluoromethane using methane and fluorine gas?**

No, not in practice. Fluorine is the most reactive halogen and reacts with methane violently (explosively), even in the dark. The reaction cannot be controlled: it gives a mixture of polyfluorinated products, carbon and HF instead of pure fluoromethane.

The textbook gives halogenation of alkanes only for chlorine (and bromine); the behaviour of fluorine is general knowledge.

21.2 REACTIVITY OF HALOGENOALKANES

Q.16 [Long] **Explain the reactivity of halogenoalkanes on the basis of bond energies.**

A halogenoalkane molecule (R–X) consists of an alkyl group with a partial positive charge on the carbon attached to the halogen, and the halogen atom with a partial negative charge. Experiments show that the **strength of the carbon-halogen bond** is the main factor that decides the reactivity: the weaker bond is more easily broken and makes the molecule more reactive.

Bond	C-F	C-Cl	C-Br	C-I
Bond energy (kJ/mol)	467	346	290	228

So iodo compounds (weakest bond) are the most reactive and fluoro compounds the least reactive:

$$R–I > R–Br > R–Cl > R–F$$

The C–F bond is so strong that fluoroalkanes do not react under ordinary conditions. Other factors that may affect the reactivity are bond polarity, steric hindrance, inductive effect, etc.

Q.17 [Short] Why do fluoroalkanes not react under ordinary conditions?

The C–F bond is very strong (467 kJ/mol), so it is not easily broken. Fluoroalkanes are therefore the least reactive halogenoalkanes.

21.3 REACTIONS OF HALOGENOALKANES

Q.18 [Short] Why can halogenoalkanes undergo both nucleophilic substitution and elimination reactions?

The C–X bond is polar due to the difference of electronegativity between C and the halogen. The carbon of the C–X bond acts as an electrophilic centre, which can be attacked by a nucleophile. Alternatively, a hydrogen attached to the adjacent carbon atom may be attacked by a base. This gives two types of reaction:

- **Nucleophilic substitution (S_N) reactions:** the halogen is replaced by some other atom or group. $Nu^- + R–X \rightarrow Nu–R + X^-$ (X^- is the leaving group).
- **Elimination (E) reactions:** HX is removed from the halogenoalkane, forming an alkene.

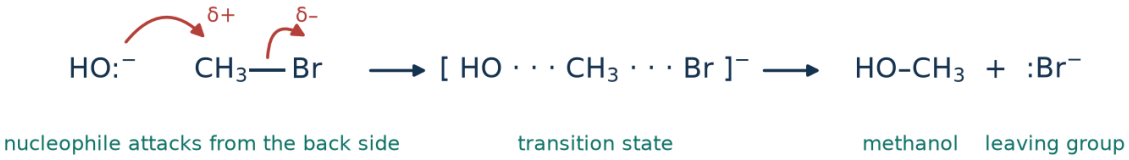
21.3.1 Nucleophilic Substitution Reactions

Q.19 [Short] What is the basic difference between the S_N2 and S_N1 mechanisms?

Nucleophilic substitution on alkyl halides involves two main processes: the departure of the leaving group (breaking of the C–X bond) and the attack of the nucleophile (formation of the C–Nu bond). If the two processes occur simultaneously in a single step, the mechanism is **S_N2**. If the bond breaks in a first step, followed by formation of the new bond in a second step, the mechanism is **S_N1**.

Q.20 [Long] Explain the S_N2 mechanism with an example and its kinetics.

Nucleophilic Substitution Bimolecular (S_N2) is a single-step mechanism. As soon as the nucleophile starts attacking the electrophilic carbon of the substrate, the bond to the leaving group starts breaking: the extent of bond formation is equal to the extent of bond breaking. The nucleophile attacks from the side **opposite** to the leaving group. Everything occurs at the same time, through a transition state.



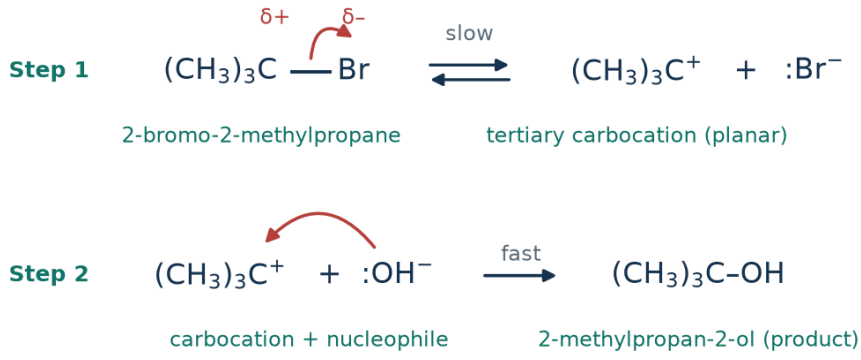
Since the reaction takes place in only one step, which is also the rate-determining step, and two molecules take part in it, it is a bimolecular reaction. The rate depends on the concentrations of both the alkyl halide and the nucleophile (second order):

$$\text{Rate} = k\,[\text{Alkyl halide}]^1\,[\text{Nucleophile}]^1 \text{ order} = 1 + 1 = 2$$

Primary monohalogenoalkanes always follow the S_N2 mechanism. This is due to steric reasons: the substrate carbon is attached to hydrogen atoms or only one carbon atom, so the nucleophile can approach and attack the electrophilic carbon easily.

Q.21 [Long] Explain the S_N1 mechanism with an example and its kinetics.

Nucleophilic Substitution Unimolecular (S_N1) involves two steps. The first step is the reversible ionization of the alkyl halide in the presence of aqueous acetone or aqueous ethyl alcohol, giving a **carbocation** as an intermediate. In the second step, the carbocation is attacked by the nucleophile to give the product.



The first step involves breaking a covalent bond, so it is slow; the second step is the energetically favourable combination of ions, so it is fast. The first step is therefore the rate-determining step. Only one molecule takes part in it, so the mechanism is unimolecular and shows first order kinetics:

$$\text{Rate} = k\,[\text{Alkyl halide}]$$

Tertiary monohalogenoalkanes always follow the S_N1 mechanism. The major reason is the formation of the stable tertiary carbocation. Moreover, three bulky groups attached to the electrophilic carbon prevent the nucleophile from attacking from the back side of the leaving group. A **secondary** halogenoalkane can undergo both S_N2 and S_N1, depending on the nature of the solvent.

Q.22 [Short] Give the order of stability of carbocations with reason.

A carbocation is formed in the rate-determining step, so the more stable the carbocation, the more easily it is formed. Stability decreases as the positive inductive effect (+I) of the alkyl groups decreases:

$$(CH_3)_3C^+ > (CH_3)_2CH^+ > CH_3CH_2^+ > CH_3^+$$

tertiary > secondary > primary > methyl

Q.23 [Short] Differentiate between S_N2 and S_N1 reactions.

S_N2	S_N1
Single step; bond making and bond breaking occur together.	Two steps; ionization first, then attack of the nucleophile.
Passes through a transition state; no intermediate.	A carbocation intermediate is formed.
Bimolecular; second order: Rate = k[RX][Nu].	Unimolecular; first order: Rate = k[RX].
Nucleophile attacks from the side opposite to the leaving group.	Nucleophile attacks the carbocation after the leaving group has gone.
Followed by primary halogenoalkanes (least steric hindrance).	Followed by tertiary halogenoalkanes (stable carbocation, crowded carbon).

Q.24 [Short] What is meant by the rate-determining step?

It is the slowest step of a reaction mechanism, which decides the overall rate. In S_N1 it is the first step, the ionization of the alkyl halide, because a covalent bond is broken.

21.3.2 β-Elimination Reactions

Q.25 [Short] What is a β-elimination reaction?

β-Elimination is a reaction in which a hydrogen atom attached to the β-carbon of the alkyl halide and the halogen atom on the adjacent carbon are removed. It occurs when a base attacks the β-hydrogen atom. The halogen is removed either simultaneously with, or after, the removal of the hydrogen atom, so an alkene is obtained instead of a substitution product. These reactions take place simultaneously with substitution reactions and often compete with them. They follow two mechanisms: E2 or E1.

Q.26 [Long] Explain the E2 mechanism.

Elimination Bimolecular (E2): the base attacks the β-hydrogen atom and the leaving group leaves at the same time, with the formation of a carbon–carbon double bond, in a single step.

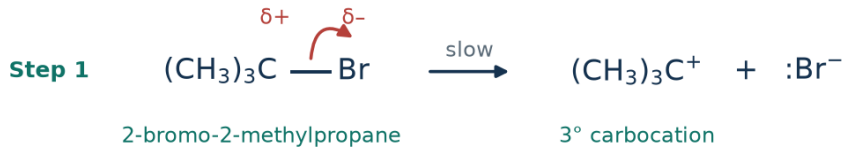


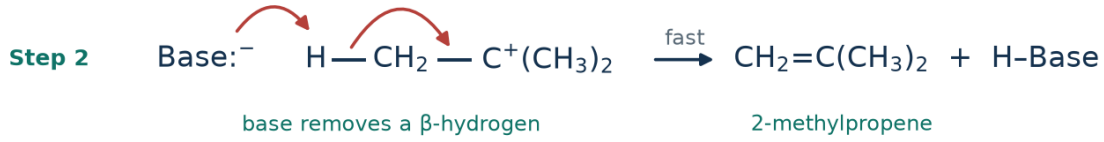
The rate depends on the concentrations of both the alkyl halide and the base; the molecularity is two and the reaction shows second order kinetics:

$$\text{Rate} = k\,[\text{Alkyl halide}]^1\,[\text{Base}]^1 \text{ order} = 1 + 1 = 2$$

Q.27 [Long] Explain the E1 mechanism.

Elimination Unimolecular (E1): like S_N1, the first step is the slow ionization of the substrate to give a carbocation. In the second step, the base attacks a β-hydrogen atom to give an alkene.





E1 is unimolecular: the rate depends only on the concentration of the alkyl halide (Rate = k [Alkyl halide]). Primary halogenoalkanes generally follow E2, tertiary halogenoalkanes follow E1, and secondary halogenoalkanes can show both, according to the choice of solvent.

Q.28 [Short] Differentiate between E2 and E1 reactions.

E2	E1
Single step.	Two steps.
Base removes the β-hydrogen and the leaving group leaves at the same time.	Slow ionization gives a carbocation; then the base removes a β-hydrogen (fast).
No intermediate.	Carbocation intermediate.
Bimolecular; second order: Rate = k[Alkyl halide][Base].	Unimolecular; first order: Rate = k[Alkyl halide].
Generally followed by primary halogenoalkanes.	Followed by tertiary halogenoalkanes.

E2	E1
Example: bromoethane → ethene.	Example: 2-bromo-2-methylpropane → 2-methylpropene.

Q.29 [Short] What is common between the S_N1 and E1 mechanisms?

In both, the first step is the same: the slow ionization of the alkyl halide to give a carbocation. Both are unimolecular and first order. They differ in the second step: the nucleophile attacks the carbocation in S_N1, while a base removes a β-hydrogen in E1.

21.4 EXAMPLES OF REACTIONS OF HALOGENOALKANES

Q.30 [Long] Describe the reactions of a halogenoalkane with sodium hydroxide.

A halogenoalkane reacts with sodium hydroxide to give both substitution and elimination products, depending on the conditions. The hydroxide ion is both a good nucleophile and a good base. As a nucleophile it replaces the halogen atom to form an alcohol; as a base it removes the β-hydrogen and the halogen to give an alkene. The reaction is carried out at higher temperature under reflux, to avoid the loss of volatile products.

$$CH_3–CH_2–Br + NaOH \rightarrow CH_3–CH_2–OH + NaBr \text{ (aqueous; substitution)}$$
$$CH_3–CH_2–Br + NaOH \rightarrow CH_2=CH_2 + H_2O + NaBr \text{ (alcohol; elimination)}$$

Q.31 [Short] Why is the reaction with sodium hydroxide carried out under reflux?

The reaction takes place at a higher temperature. Reflux conditions are used to avoid the loss of volatile products (and reactants) during heating.

Q.32 [Short] Describe the reaction of halogenoalkanes with KCN.

The cyanide ion (CN^-) is a strong nucleophile but a weak base, so it gives the substitution product, a nitrile. The halogenoalkane is heated under reflux with an ethanolic solution of potassium cyanide. Bromoethane gives propanenitrile (the chain becomes one carbon longer).

$$CH_3–CH_2–Br + KCN \rightarrow CH_3–CH_2–C\equiv N + KBr \text{ (alcohol, reflux)}$$

Q.33 [Long] Describe the reactions of halogenoalkanes with ammonia.

Ammonia is a good nucleophile and reacts with halogenoalkanes favourably. Different types of amines can be produced if the amounts of halogenoalkane and ammonia are carefully controlled.



Ethanamine is also a nucleophile, so it competes with ammonia for the alkyl halide and the reaction continues:

$$NH_3 + C_2H_5Br \rightarrow C_2H_5NH_2 + HBr \text{ (ethanamine)}$$
$$C_2H_5NH_2 + C_2H_5Br \rightarrow (C_2H_5)_2NH + HBr \text{ (N-ethylethanamine)}$$
$$(C_2H_5)_2NH + C_2H_5Br \rightarrow (C_2H_5)_3N + HBr \text{ (N,N-diethylethanamine)}$$
$$(C_2H_5)_3N + C_2H_5Br \rightarrow [(C_2H_5)_4N]^+Br^- \text{ (quaternary ammonium salt)}$$

If a large excess of bromoethane is used, the final product is the quaternary ammonium salt. (The learning outcome states the conditions: NH_3 in ethanol, heated under pressure.)

The textbook names the final salt "N,N,N-triethylethanaminebromide"; it is also called tetraethylammonium bromide.

Q.34 [Long] How is aqueous silver nitrate used to identify the halogen present in a halogenoalkane?

The reaction is carried out in a mixture of water and ethanol; the alcohol helps to dissolve the halogenoalkane. Water acts as a nucleophile and replaces the halide ion:

$$CH_3–CH_2–X + H–OH \rightarrow CH_3–CH_2–OH + H^+ + X^-$$

The released halide ion reacts with silver nitrate to give a precipitate with a characteristic colour, which identifies the halogen present:

$$Ag^+ + Cl^- \rightarrow AgCl \text{ (white ppt)}$$
$$Ag^+ + Br^- \rightarrow AgBr \text{ (cream yellow ppt)}$$
$$Ag^+ + I^- \rightarrow AgI \text{ (bright yellow ppt)}$$

The rate of formation of the precipitate shows the reactivity: the quicker the precipitate appears, the more reactive the halogenoalkane. An iodoalkane gives a precipitate much more quickly than a bromoalkane; a chloroalkane reacts very slowly.

Q.35 [Short] Why is ethanol added in the silver nitrate test?

The reaction is carried out in a mixture of water and ethanol. The halogenoalkane does not dissolve in water; the alcohol helps to dissolve it so that it can react with water.

Quick Check 21.3

Q.36 [Short] **a) Why halogenoalkanes are considered as very reactive class of organic compounds?**

The C–X bond is polar because the halogen is more electronegative than carbon. The carbon carries a partial positive charge and acts as an electrophilic centre that is easily attacked by nucleophiles, and the halide ion is a good leaving group. A base can also remove the hydrogen from the adjacent carbon. So halogenoalkanes readily undergo both nucleophilic substitution and elimination reactions.

Q.37 [Short] **b) Complete the following equation. $C_2H_5Br + CH_3COONa \rightarrow$**

$$C_2H_5Br + CH_3COONa \rightarrow CH_3COOC_2H_5 + NaBr$$

The ethanoate ion (CH_3COO^-) is the nucleophile; it replaces bromine to give the ester ethyl ethanoate.

Q.38 [Short] **c) How will you convert ethane into ethanol and ethene?**

First convert ethane into chloroethane; then use sodium hydroxide under two different conditions.

$$CH_3–CH_3 + Cl_2 \rightarrow CH_3–CH_2–Cl + HCl \text{ (light)}$$
$$CH_3–CH_2–Cl + NaOH \rightarrow CH_3–CH_2–OH + NaCl \text{ (aqueous: ethanol)}$$
$$CH_3–CH_2–Cl + NaOH \rightarrow CH_2=CH_2 + NaCl + H_2O \text{ (alcohol: ethene)}$$

Q.39 [Short] **d) Give the reaction of bromopropane with KCN. Is this a substitution reaction or an elimination reaction? Name the main product of this reaction.**

$$CH_3–CH_2–CH_2–Br + KCN \rightarrow CH_3–CH_2–CH_2–C{\equiv}N + KBr \text{ (alcohol, reflux)}$$

It is a **nucleophilic substitution** reaction, because CN^- is a strong nucleophile but a weak base. The main product is **butanenitrile**.

21.5 HALOGENOARENES

Q.40 [Short] **How are halogenoarenes prepared?**

Halogenoarenes such as chlorobenzene and bromobenzene are prepared by reacting benzene with chlorine or bromine in the presence of a catalyst, anhydrous $FeCl_3$ or $FeBr_3$. The reaction is an electrophilic aromatic substitution and is typically carried out at room temperature.

$$C_6H_6 + X_2 \rightarrow C_6H_5X + HX \ (FeX_3)$$

21.5.1 Comparison of Reactivities of Halogenoalkanes and Halogenoarenes

Q.41 [Long] Compare the reactivity of halogenoalkanes and halogenoarenes, using chloroethane and chlorobenzene as examples.

Halogenoalkanes (chloroethane)	Halogenoarenes (chlorobenzene)
Aliphatic: the halogen is attached to an alkyl group.	Aromatic: the halogen is attached directly to the benzene ring.
The carbon attached to the halogen has a partial positive charge (electrophile) and invites nucleophiles.	The lone pair of chlorine is delocalised into the aromatic ring.
C–Cl is an ordinary single bond, easily broken.	C–Cl has partial double-bond character; it is stronger and difficult to break.
Undergo nucleophilic substitution, and elimination with bases.	Less reactive towards nucleophiles; synthetically less useful.
$CH_3CH_2Cl + NaOH_{(aq)} \rightarrow CH_3CH_2OH + NaCl$	No reaction with NaOH under normal conditions.

Q.42 [Short] Why is the C–Cl bond in chlorobenzene stronger than in chloroethane?

In chlorobenzene the electron pair on the chlorine atom is delocalised due to its involvement with the aromatic ring. This gives the carbon–halogen bond partial double-bond character, so it is stronger and difficult to break.

Quick Check 21.4

Q.43 [Short] a) What is the role of iron halide in the preparation of halogenoarene?

The iron halide (anhydrous $FeCl_3$ or $FeBr_3$) is a catalyst. It acts as a Lewis acid (halogen carrier): it polarises the halogen molecule and generates the electrophile X^+ ($X_2 + FeX_3 \rightarrow X^+ + [FeX_4]^-$), which attacks the benzene ring. It is regenerated at the end.

Q.44 [Short] b) Chlorobenzene does not react with NaOH under normal conditions, but chloroethane does; explain why?

In chloroethane the carbon attached to chlorine carries a partial positive charge and the C–Cl bond is an ordinary single bond, so the hydroxide ion attacks this carbon and replaces chlorine. In chlorobenzene the lone pair of chlorine is delocalised into the aromatic ring, which gives the C–Cl bond partial double-bond character. The bond is stronger and difficult to break, so chlorobenzene does not react under normal conditions.

Q.45 [Short] c) Can chlorobenzene undergo elimination? Explain why or why not?

No, not under normal conditions. The C–Cl bond is strong because of its partial double-bond character, so chloride does not leave easily. Also, removing H and Cl from two adjacent ring carbons would require a triple bond inside the six-membered ring, which would be highly strained and would disturb the stable aromatic system.

The textbook only states that the strong C–Cl bond makes halogenoarenes less reactive; the point about the ring is standard reasoning.

Q.46 [Short] Why has the use of DDT been restricted, and what problem has this caused?

Some halogen compounds like DDT have been used as pesticides, but their persistence in the environment has led to restrictions or banning of their use in developed countries. Because DDT is a cheap and effective mosquito control agent, some developing countries have experienced a sharp increase in malaria deaths after its removal from stores. It is therefore advised to restrict its use for environmental protection.

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